

2-24

$$\begin{aligned} (1) \quad & \frac{1}{2} u[n] \rightarrow y_1[n] \\ & \frac{1}{2} u[n] \rightarrow y_2[n] \\ & \frac{1}{2} u[n] \rightarrow y_3[n] - y_1[n] \\ & 2y_1[n] - y_2[n] \\ & = -2u[n] \\ & \text{零输入响应 } -2u[n] \end{aligned}$$

(2)

$$\begin{aligned} & x_1[n] \text{ 对应的零状态} \\ & \text{响应 } y_1[n] + 2u[n] \\ & = 2 \cdot \frac{1}{2} u[n] - \frac{1}{2} u[n] \\ & x_2[n] \rightarrow \frac{1}{2} u[n] - \frac{1}{2} \left(\frac{1}{2} \right) u[n] \\ & \quad - 2u[n] \\ & = -\frac{1}{2} \left(\frac{1}{2} \right) u[n] \end{aligned}$$

$$\begin{aligned} & \text{完全响应是 } -\frac{1}{2} u[n] \\ & k=0 \text{ 时 } a_k=0 \\ & k \neq 0 \text{ 时 } a_k = \frac{E}{\omega} \sin(k\pi) \end{aligned}$$

3-1

$$\begin{aligned} (1) \quad & y(t) = H(j\omega) e^{j\omega t} \\ & x(t) = E \sin \omega t \rightarrow y(t) = E H(j\omega) = E |H(j\omega)| \cos(\omega t + \angle H(j\omega)) \end{aligned}$$

$$\begin{aligned} (2) \quad & y(t) = \sum_{k=-\infty}^{\infty} a_k H(jk\omega) e^{jk\omega t} \\ & = \sum_{k=-\infty}^{\infty} a_k \frac{1}{j k \omega} e^{jk\omega t} \\ & \quad + \sum_{k=0}^{\infty} a_k \left(\frac{k}{5} + 1 \right) e^{jk\omega t} \end{aligned}$$

$$\begin{aligned} & \sum_{k=-\infty}^{\infty} a_k \left(\frac{k}{5} + 1 \right) e^{jk\omega t} \\ & = \sum_{k=-\infty}^{\infty} a_k \left(\frac{k}{5} + 1 \right) e^{jk\omega t} \end{aligned}$$

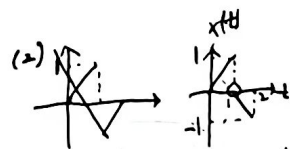
$$\begin{aligned} & + \sum_{k=0}^{\infty} a_k \left(\frac{k}{5} + 1 \right) e^{jk\omega t} \\ & y(t) = \sum_{k=-\infty}^{\infty} a_k \left(\frac{k}{5} + 1 \right) e^{jk\omega t} + \sum_{k=0}^{\infty} a_k \left(\frac{k}{5} + 1 \right) e^{jk\omega t} \end{aligned}$$

3-4

$$\begin{aligned} (1) \quad & x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega t} \quad \omega = \frac{2\pi}{T} \\ & x(t-T) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega(t-T)} \\ & = \sum_{k=-\infty}^{\infty} (-1)^k a_k e^{jk\omega t} \end{aligned}$$

$$\begin{aligned} & x(t) + x(t-T) = 0 \\ & (1 + (-1)^k) a_k = 0 \\ & k \text{ 为偶数时 } a_k = 0 \end{aligned}$$

$x(t)$ 傅里叶级数只有奇次分量



$$\begin{aligned} & a_k = \frac{1}{2} \left\{ \frac{1 - (-1)^k}{j k \pi} + 2 \frac{[(-1)^k - 1]}{(k\pi)^2} \right\} \\ & k \text{ 偶数 } a_k = 0 \\ & k \text{ 奇数 } a_k = \frac{1}{j k \pi} - \frac{2}{(k\pi)^2} \end{aligned}$$

(1)

请编写MATLAB程序

`H = REVERSECONV(X, Y)`

要求输入序列X和Y，输出序列H，使 $Y=X*H$ ，若没有H满足要求，则输出错误信息“No sequence satisfies this condition.”

```

function H = reverseconv(X,Y)
errMsg = 'No sequence satisfies this condition.';

% Check the validity of inputs.
if ~isvector(X) || ~isvector(Y) || isempty(X) || isempty(Y)
    error(errMsg);
end

X = X(:).';
Y = Y(:).';

% The length of H is length(Y) - length(X) + 1.
if length(Y) < length(X)
    error(errMsg);
end

% The first element of X must be nonzero to ensure the uniqueness of H.
if X(1) == 0
    error(errMsg);
end

% Solve H recursively from Y = conv(X, H).
H = zeros(1, length(Y) - length(X) + 1);
Xpad = [X, zeros(1, length(H) - 1)];

H(1) = Y(1) / X(1);
for i = 2:length(H)
    H(i) = (Y(i) - H(1:i-1) * Xpad(i:-1:2).') / X(1);
end

if any(abs(conv(X, H) - Y) > 1e-8)
    error(errMsg);
end
end

```

2(1)

请编写程序，画出当 $T=50$ 时的 $h_T(t)$ 的波形，其中 $t \in (-20, 20)$ 。

```

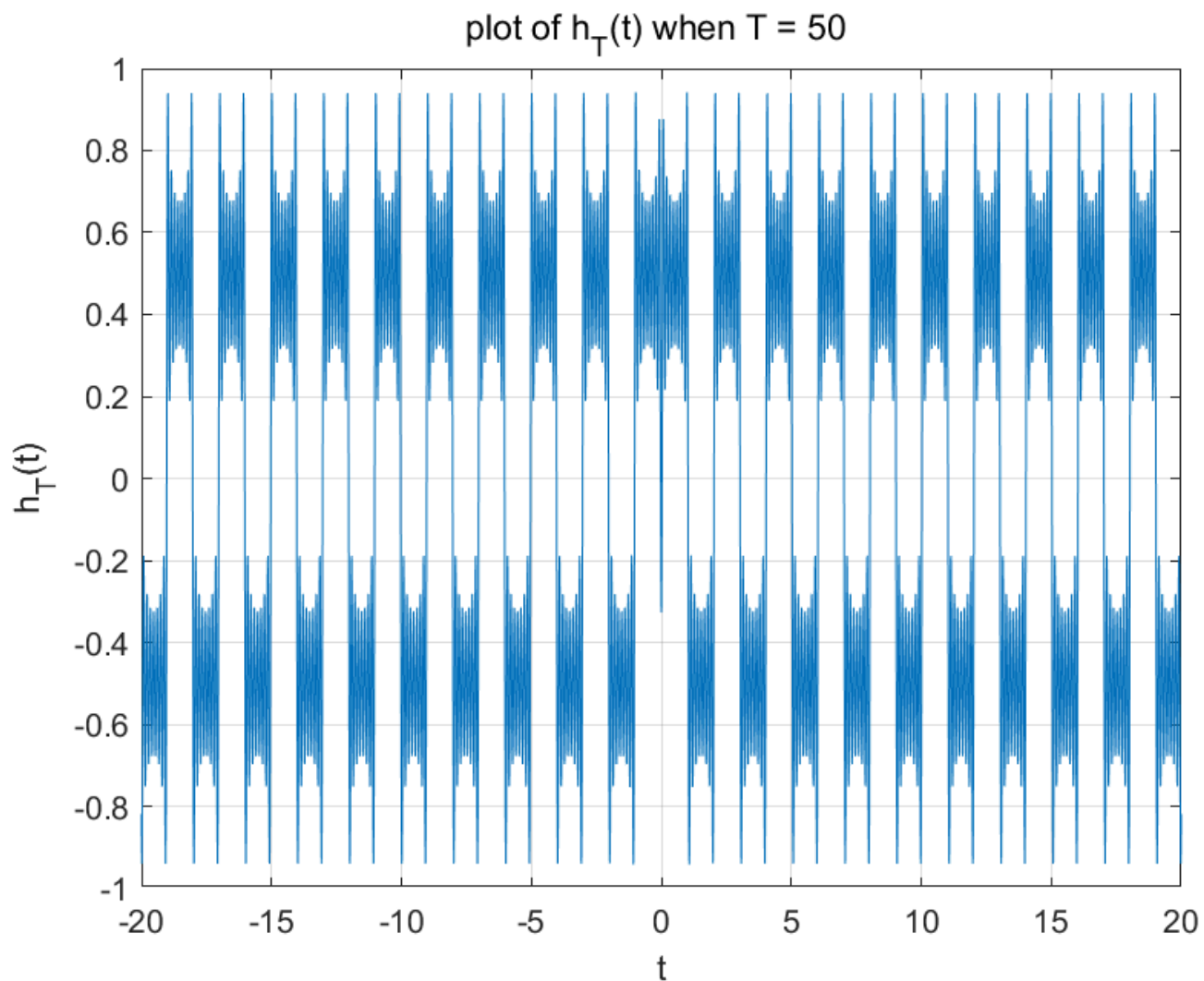
t = linspace(-20,20,40001);
T = 50;
dt = t(2) - t(1); % 采样间隔
w = linspace(0, T, 50000); % 频率范围从0到T, 分成1000个点
y = @(w,x) sin(0.5*w).^2./(w.*sin(w)).*cos(w*x);
h = zeros(1,length(t));

for i = 1:length(t)
    for j = 1:length(w)
        if w(j) == 0
            y_val = 0.25 * cos(0); % 当 w=0 时, 使用极限值
        else
            if abs(sin(w(j))) < 1e-6 %奇点跳过
                continue;
            else
                y_val = y(w(j), t(i));
            end
        end
        h(i) = h(i) + y_val * (w(2)-w(1)); % 使用矩形法则进行数值积分
    end
end
h = h*2/pi;

plot(t,h);
xlabel('t');
ylabel('h_T(t)');
title('plot of h_T(t) when T = 50 ');
grid on;
saveas(gcf,"h_T(t).png");

```

结果如下图所示



可以看到 $h(t)$ 是无限震荡的函数。

2(2)

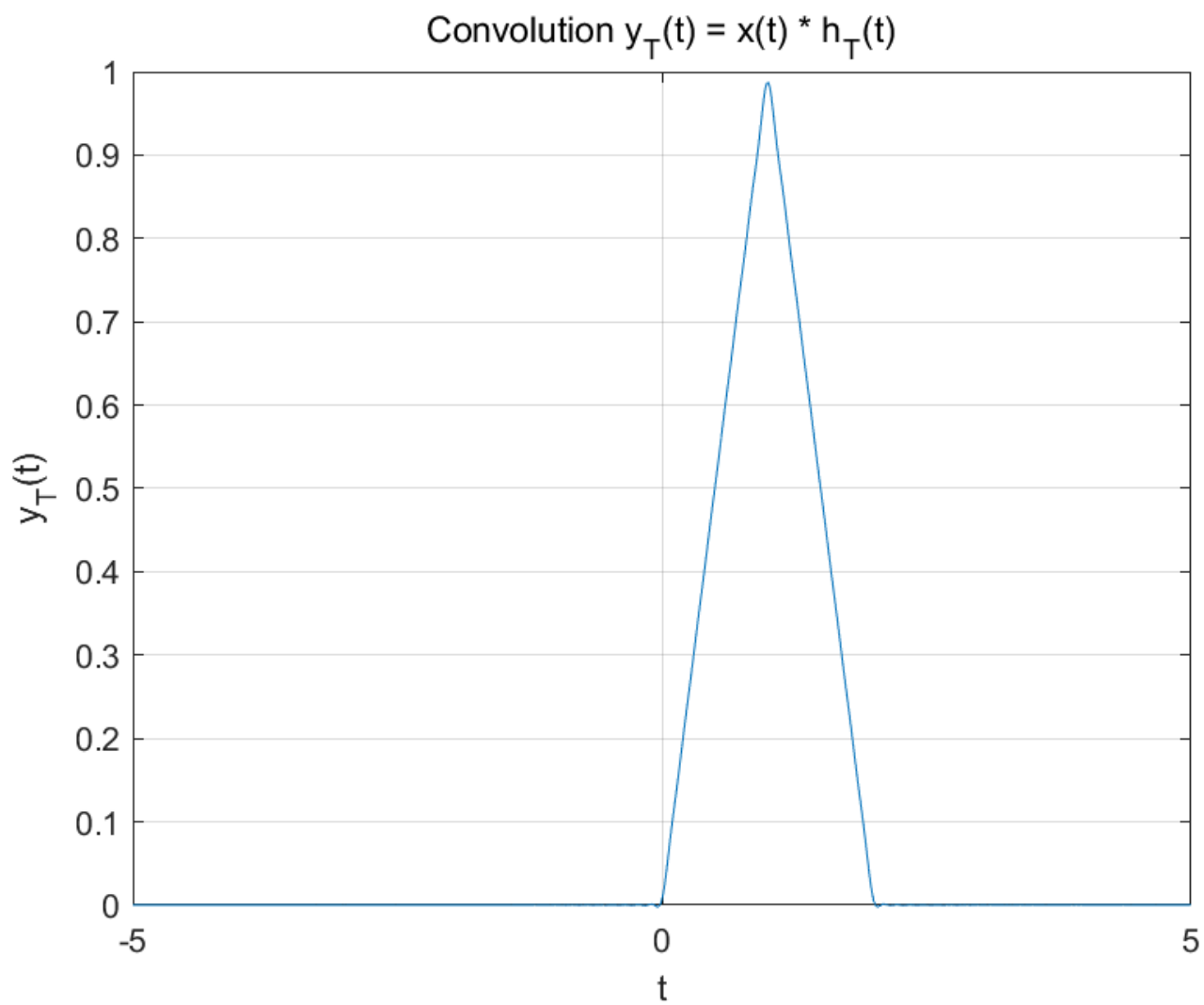
基于以上 $h_T(t)$ ，利用卷积公式，画出当 $T=50$ 时 $y_T(t) = x(t) * h_T(t) = \int_{-T}^T x(\tau) h_T(t - \tau) d\tau$ 的波形，其中 $t \in (-5, 5)$ 。看一下 $y_T(t)$ 是不是一个近似的三角波。

首先根据 $x(t)$ 的定义化简积分表达式，得到

$$y_T(t) = \int_0^2 h_T(t - \tau) d\tau$$

基于此进行计算。

结果如下图所示



可以看到 $y_T(t)$ 几乎是一个三角波。

代码如下

```

newT = -5:dt:5; % 卷积结果的时间范围和采样点
y = zeros(1,length(newT));
% Calculate the convolution

for i = 1:length(newT)
    time = newT(i);
    for tao = 0:dt:2 % tau 从0到2, 因为 x(tau) 在这个范围内为1
        index = 1+round((time - tao-t(1))/dt); % 计算 h 的索引
        if (index <= length(h)) && index > 0 % 确保索引在 h 的范围内
            y(i) = y(i) + h(index) * dt; % 使用矩形法则进行数值积分
        end
    end
end

% Plot y_T(t)
figure; % Create a new figure
plot(newT, y);
xlabel('t');
ylabel('y_T(t)');
title('Convolution y_T(t) = x(t) * h_T(t)');
grid on;
saveas(gcf, 'y_T(t).png');

```